88/100

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Short Answer Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 5

Duration: 60 Minutes **Total Marks: 25**

Code of Conduct

I M. Gopi Krishna Reddy. 192372055 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _______

Criteria	Conceptual Accuracy (1.5)	Coverage of Concepts (1)	Use of Examples (0.75)	Comparison / Differentiation(1)	Clarity of Expression(0.75)	TOTAL (5)
Weightage	30%	20%	15%	20%	15%	100%
Q1						
Q2						
Q3						
Q4						
Q5						

Signature of the Faculty **Total Marks Awarded:**

Assessment Tasks

Technical Presentation Topics

Prepare a 6-8 minute presentation on any one of the following topics. Your slides must include definitions, architecture, industry relevance, and one practical example.

1. Describing the Role of Data Centers in Cloud Computing - Interpret how data centers support cloud services.
2. Explaining the Role of Web Applications in Cloud Computing - Interpret how cloud-based applications function over the internet.
3. Describing the Role of Web Browsers in Accessing Cloud Services - Interpret how browsers act as client interfaces.
4. Understanding Cloud Storage Mechanisms - Explain how data is stored, managed, and accessed in the cloud.
5. Describing the Role of Networking in Cloud Computing - Interpret how network infrastructure supports cloud services.

Minimum slide flow:

- Title and objective
- Core technical explanation
- Architecture or process diagram
- Real-world use case
- Conclusion and key takeaway

Technical Presentation Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Technical Content	30%	Content is highly accurate, relevant, and insightful	Content is correct with minor gaps	Basic content with limited depth	Content is inaccurate or superficial
Structure and Flow	20%	Excellent logical flow and slide organization	Good structure with minor issues	Some organization but weak flow	Disorganized presentation
Use of Visuals	15%	Visuals strongly support technical communication	Relevant visuals used well	Limited or unclear visuals	No meaningful visuals
Communication Skills	20%	Confident, clear, and engaging delivery	Clear delivery with minor hesitation	Moderate clarity but uneven delivery	Weak delivery and low clarity
Professionalism	15%	Well-prepared and audience-focused presentation	Mostly professional	Basic preparation visible	Poor preparation and professionalism

1. Describing the role of data centres in cloud computing:-

The foundational Infrastructure: Interpreting the role of physical data centres in supporting cloudservices.

Objective:- To interpret and evaluate how physical data centres facilities act as the structural backbone of cloud computing, transforming raw hardware components into scalable, high availablity virtual services.

Core Explaination:- A data center is a centralized physical facility that houses, enterprise-grade computing resources including multi core processing servers, high density storage arrays, network switch, and crucial utility systems. In cloud computing, these facilities support services through three primary mechanisms:

Hardware visualization:- Physical infrastructure is abstracted using hypervisors, slicing single bare-metal servers into thousands of virtual machines to enable resource pooling.

Environmental and power management:- Data centres feature industrial HVAC cooling systems, uninterruptible powers supplies and disel generators to protect hardware from overheating.

Geographic zones and availability:- Cloud facilities are organized into seperate isolated regions to allow automated data mirroring reducing latency and securing disaster recovery pathways.

Architecture:-

[User Application Request] → [Virtualization Control layer] → [Physical facility Infrastructure]

User Application Request	Virtualization Control layer	Physical facility Infrastructure
↓	↓	↓
Routed over Internet/VPN	Hypervisors allocate CPU, RAM & Storage	Server Racks ↓ Power/Cooling

Real-world Use Caser

Case studyr AWS (Amazon Web Service)

Implementationr AWS pools multiple physical data centres into distinct "Availability zones" within a geographic region. When an enterprice deployed a platform, the underlying data center mechanisms automatically spit the virtual workload accross physical buildings.

conclusionr physical data Centres represents the necessary tangable base of cloud systems. Converting raw hardware into elastic utility compouting models

2. <u>Explaining the role of web applications in cloud computing</u>r

Decoupled Architecturer explaining the function and delivery of cloud based applications.

Objectiver To interpret the internal architecture of cloud based applications and explain how they execute businees logic over the internet without using local device storage or processing power.

Core explainationr cloud web applications are Software programs where the primary application logic, computation and database storage reside on remote cloud servers rather than the local endpoint device. Their performance over the internet depends on these core pillars.

Client-Server Decoupling:r The architecture seperates the lightweight front-end User interface from the cloud-hosted backend processing environment.

API-Driven Communicationr Systems use lightweights protocols to exchange structured JSON data packets quickly over.

Microservices Design:- Applications are broken down into independ-ent functional modules running inside isolated cloud Containers that Scale automatically to match live user traffic.

Architecture:-

(Client Interface/ Device UI) — (HTTPS API Request) → [Public Internet]
↓
[Managed Databases] ↔ [Microservices Logic Tier] ← [cloud API Gate way]

Real-world Use Case:- when a user updates a document the local machine only handles rendering the graphical interface Every text entry. format modification or Collaboration event is transm-itted instantly as an network packet to Google's backend cloud backend handles the real-time coauthoring synchronization and save changes directly to managed cloud database stores

Conclusion:- Cloud-based web applications eliminate device-specific hardware dependencies by offloading processing loads entirely onto scalable remote cloud servers.

3. Describing the Role of web Browsers in Accessing cloud Services:-

The Universal Interface:- Evaluating the Role of web Browsers as client Gateways to cloud services.

Objective:- To analyze how modern web browsers function as standardized lightweight client interfaces (thin clients) that transclate distant Cloud data into rich.

Core Explaination:- A web browser acts as the endpoint applica-tion layer that handles Communication between end-Users and Com-plex cloud Systems. It Servers as a client gateway through these primary mechanisms:

Standard Engine Rendering: Browsers use core layouts to interpret HTML, CSS and code, ensuring uniform application display accross different operating systems.

Local client Script execution: High-Speed Javascript interpreters executes interface adjustments and local interface logic directly on the User's device, reducing Server strain.

Security protocol validation: Browsers manage encryption by validating SSL/TLS certificates. preventing data interception between the client machine and the cloud host.

Architecture:

[web Browser window] → Parsed DOM UI ⟶ Local JS Engine (V8/Spidermon-key)

↕ (secure SSL/TLS Session via Internet)

[Cloud web Server Tier] → Dynamic Assets, HTML5 Templates and API payloads

Real world Use-Case: Instead of Corporate Sales department installin-g heavy data base management Software on thousand unique employee Computers Users login Standard web browsers. The brows-er Secures HTTPS network Session. runs the visual dashboard.

Conclusion: web browsers eliminate the need for installing hea-vy desktop applications by standardizing how Users Connect to cloud Software over the internet.

Con

4 Understanding cloud Storage Mechanisms:-

Software-Defined Data Systems:- Analyzing Architectures and cloud Storage Mechanisms.

Objective:- To explain the foundational mechanisms behind cloud Storage virtualization data preservation strategies and the Structural differences between object, block and file architectures

Core Explaination:- cloud storage abstracts physical media into highly resilient, automated and horizontally Scalable Storage pools accessible over networks. It relies on these main System mechanisms

Storage pool Virtualization:- Physical disk arrays are combined into flexible logical volumes, Speparating the underlying storage medium from the Software management tools.

Automatic Replication:- Data is cloned across different Server racks and geographic regions to protect against unexpected storage media failures.

Capacity Optimization:- Deduplication Software identifies and drops duplicates chunks of information, optimizing Storage limits and preserving connection speeds.

Architecture:-

client App/ web Interface → Access requests using unique IDs

↓

Control and Management layer → Metadatalogging → Encryption

↓

Deduplication

↓

Virtualized Physical media ⟶ Distributed arrays of object, Block/ file Systems

Real-world Use Case:- Netflix relies on object Storage mechanis-ms to host its global film Catalog. Movies are saved as discrete storage objects containing raw video content linked with exten-sive custom metadata tags. Cloud automation protocols handle these objects.

Conclusion:- Cloud storage replaces traditional, rigid physical stora-ge hardware with adaptive, virtual ecosystems that adjust capacity automatically based on use.

5 Describing the Role of Networking in Cloud Computing.

The digital Arteries:- Interpreting how Network Infrastructure Suppo-rts cloud Services.

Objectives:- To interpret how Software-defined network architecture routes backbones and local balancing tools link Users securly to Virtual cloud environments.

Core Explaination:- Cloud networking encompasses the physical interconnects and virtual Controls layers that govern data movem-ent between users and cloud systems. It support Services thro-ugh these key methods.

Software-Defined Networking (SDN):- Networks decouple Configura-tion rules from the physical wires. allowing Users to build Virtual firewalls, private Subnets and Custom routing pathways via Software control panels.

Load Balancing Distribution:- Automated Virtual load balancers intercept incoming app traffic and distripute the packets evenly across server pools to eleminate proformance bottlenecks.